

POLITECNICO
MILANO 1863

DEPARTMENT OF MECHANICAL
ENGINEERING

EXERCISE CLASS 6

ARIMA

Name Surname



DIPARTIMENTO DI ECCELLENZA
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Milano

REMINDER (NON RANDOM PROCESSES)

Distributional model

(observations randomly drawn from a population):

$$X \sim iid(\mu, \sigma^2)$$

To make inference (e.g. hypothesis testing): $X \sim NID(\mu, \sigma^2)$

Time series model

$$Y = f(X, \theta) + \varepsilon, \quad \varepsilon \sim iid(0, \sigma^2)$$

To make inference: $\varepsilon \sim NID(0, \sigma^2)$

Remind:

Independence $\rightarrow$ absence of autocorrelation

Absence of autocorrelation $\rightarrow$ independence ONLY IF data are normal

REMINDER (NON RANDOM PROCESSES)

Qualitative analysis of
process data

Is it random?

- *Overall process level is constant over time?*
- *Is there a systematic pattern?*
- *The variation around the process level is constant?*

Are there outliers?

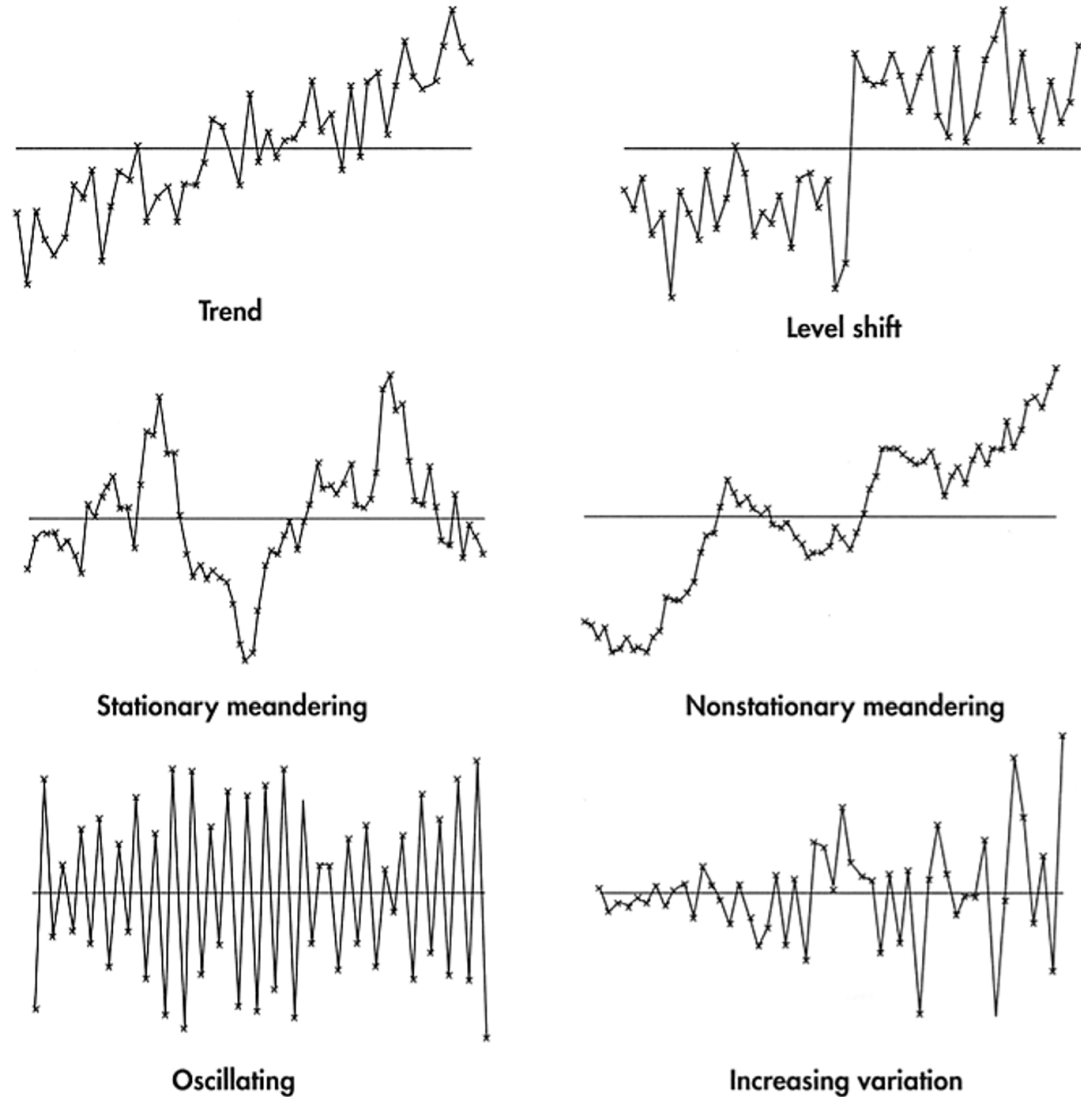


Figure 2.4 Time-series plots of different types of nonrandom process behavior.

REMINDER (previous class)

Autoregressive models: AR(p)

$$X_t = \xi + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t$$

- ACF “geometrically decays”
- PACF indicates the order p

Moving average models: MA(q)

$$X_t = \mu - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q} + \varepsilon_t$$

$$\tilde{X}_t = X_t - \mu = -\theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q} + \varepsilon_t$$

- PACF “geometrically decays”
- ACF indicates the order q

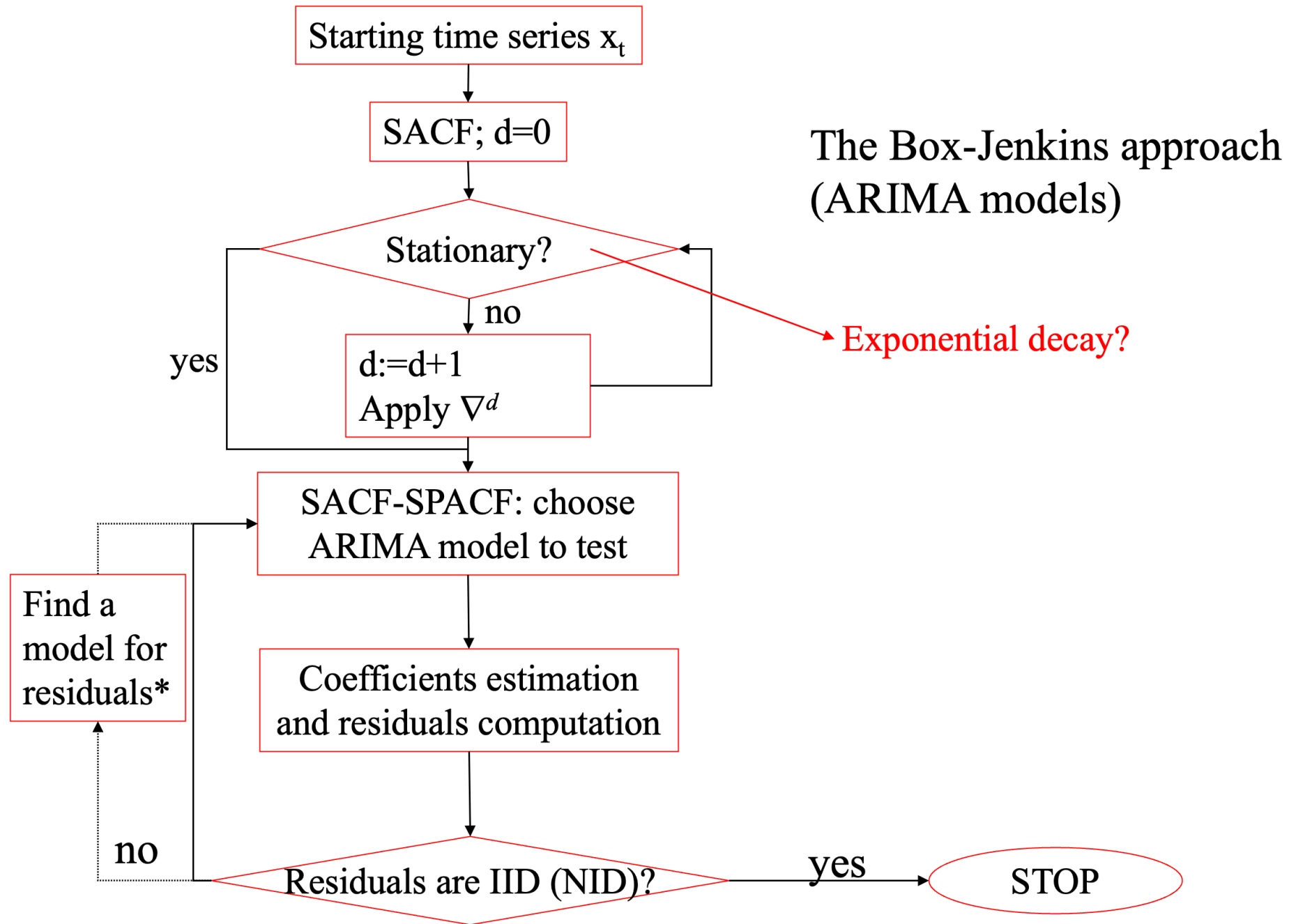
REMINDER

ARMA(p,q): both AR(p) and MA(q) terms are present

- It resembles an AR(p) after q lags
- Parsimony: low order models are preferred (easier to deal with)
- Model identification is often a *trial and error* problem

Homogeneous nonstationary ARMA (p,q) = ARIMA(p,d,q)

- ACF with slow (e.g. linear) decay, not a “geometrical decay”



Exercise 1

In a chemical process, 70 consecutive measurements were made over time for a given quality characteristic (pressure) of interest. Data are stored in `pressure.csv` (series 'F' “Time Series Analysis – 3rd edition” Box Jenkins Reinsel – Prentice Hall)

1. Estimate the most suitable ARIMA model.

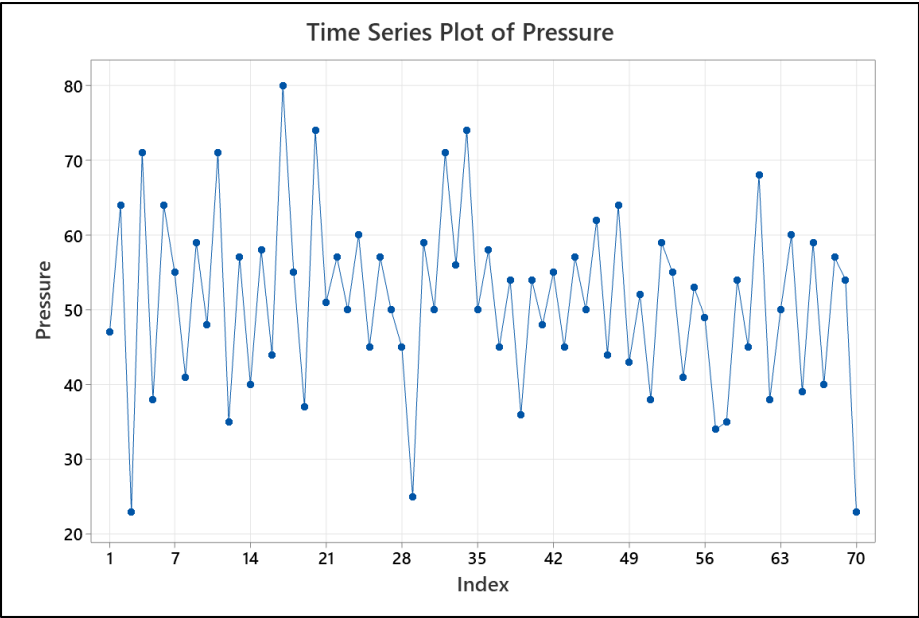
Exercise 1.1 Estimate the most suitable ARIMA model.

1) Import the data in Minitab:

File → Open → Locate “pressure.csv” and Open

2) Inspect the dataset by plotting the time series:

Graph → Time Series Plot



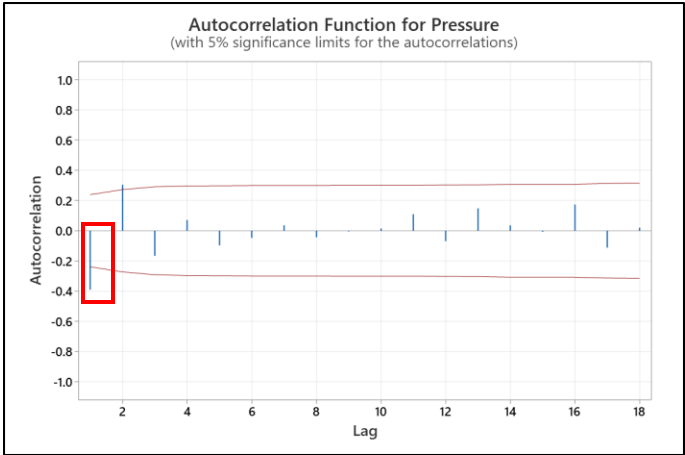
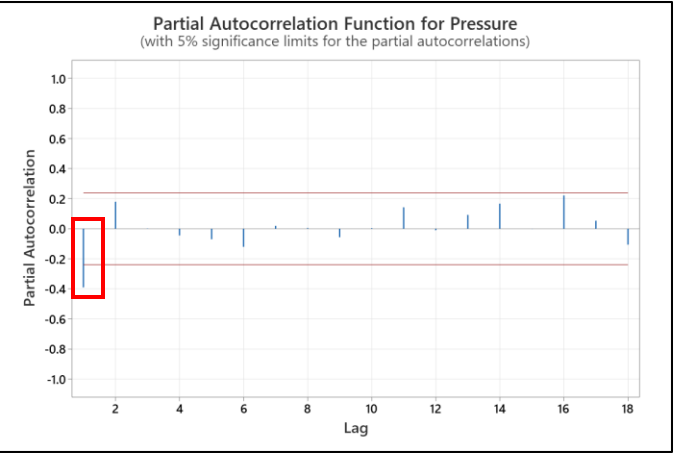
We observe a very high number of runs, a systematic variation of observations above and below the mean. This pattern is typical of negative correlated processes.

3) Check for autocorrelation:

Stats → Nonparametrics → Runs test...

Number of Runs		
Observed	Expected	P-Value
59	36.00	0.000

Stats → Time Series → ACF/PACF...



- The process looks almost stationary.
- Geometric decay of ACF?
- Most suitable model may be AR(1) with negative coefficient.

Exercise 1.1 Estimate the most suitable ARIMA model.

4) Fit an AR(1) model to the time series:

Stat → Time Series → ARIMA...

ARIMA

Series: Pressure

☐ Fit seasonal model

Period: 12

Autoregressive:

Nonseasonal

Seasonal

1

0

Difference:

0

0

Moving average:

0

0

☒ Include constant term in model

☐ Starting values for coefficients:

Select

Graphs...

Forecasts...

Results...

Storage...

Help

OK

Cancel

ARIMA: Graphs

☐ Time series plot (including optional forecasts)

Residual Plots

☒ ACF of residuals

☒ PACF of residuals

Individual plots

☐ Histogram of residuals

☐ Normal plot of residuals

☐ Residuals versus fits

☐ Residuals versus order

☒ Four in one

Residuals versus the variables:

Select

Help

OK

Cancel

ARIMA: Storage

☒ Residuals

☒ Fits

☒ Coefficients

Help

OK

Cancel

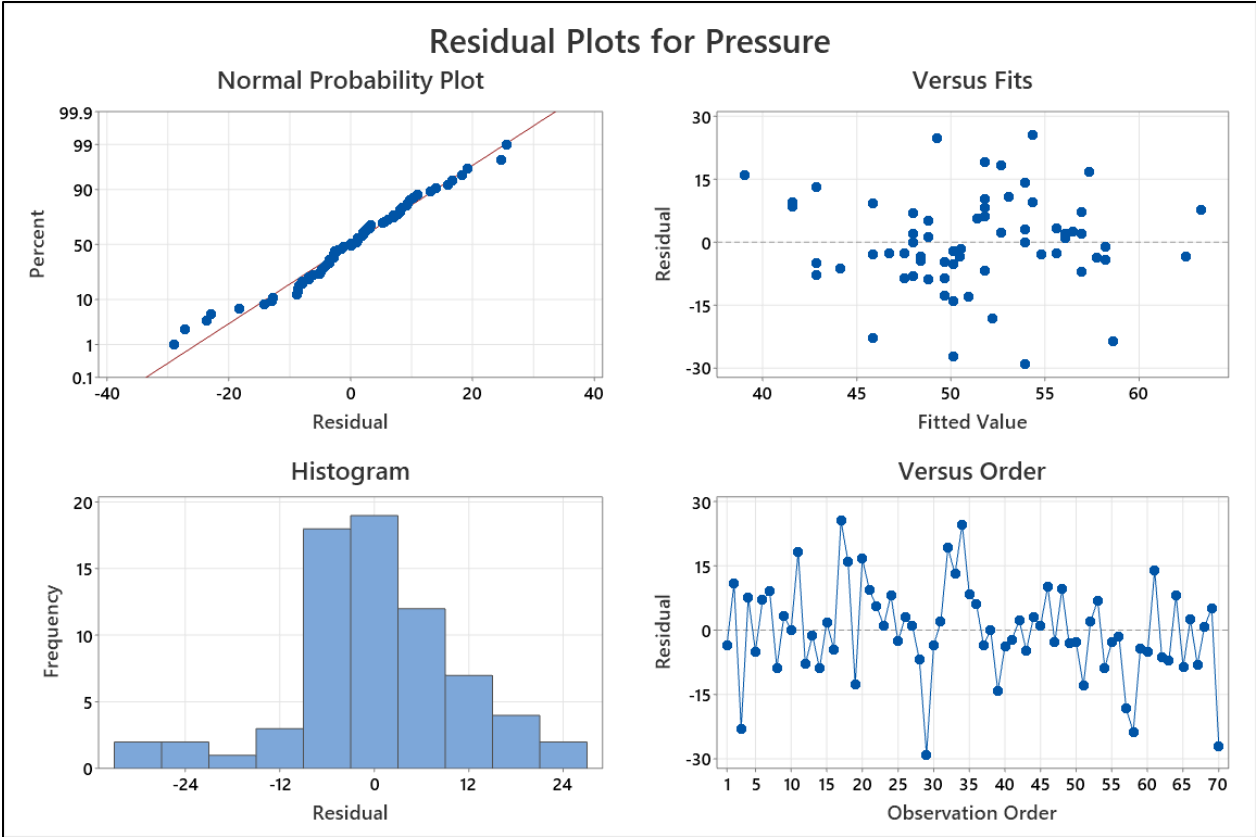
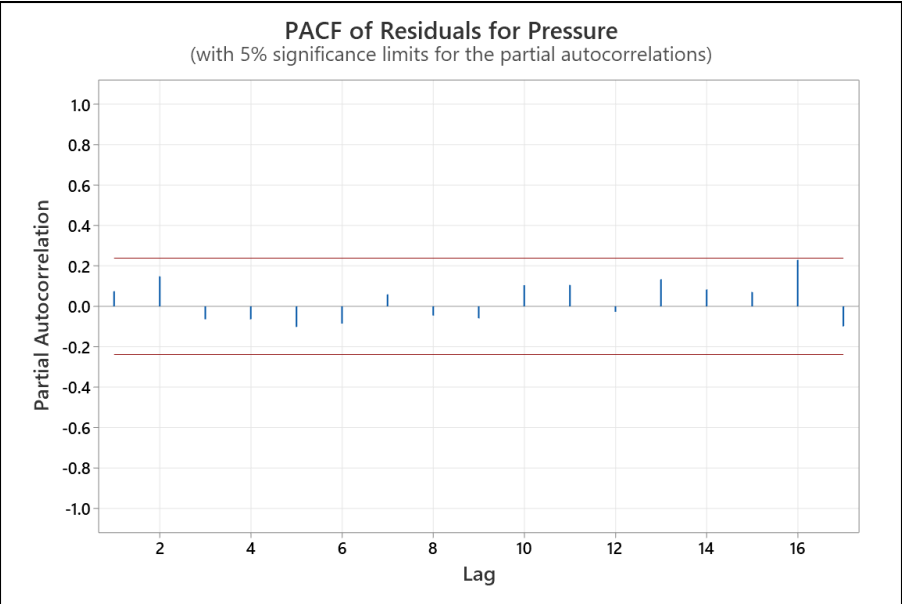
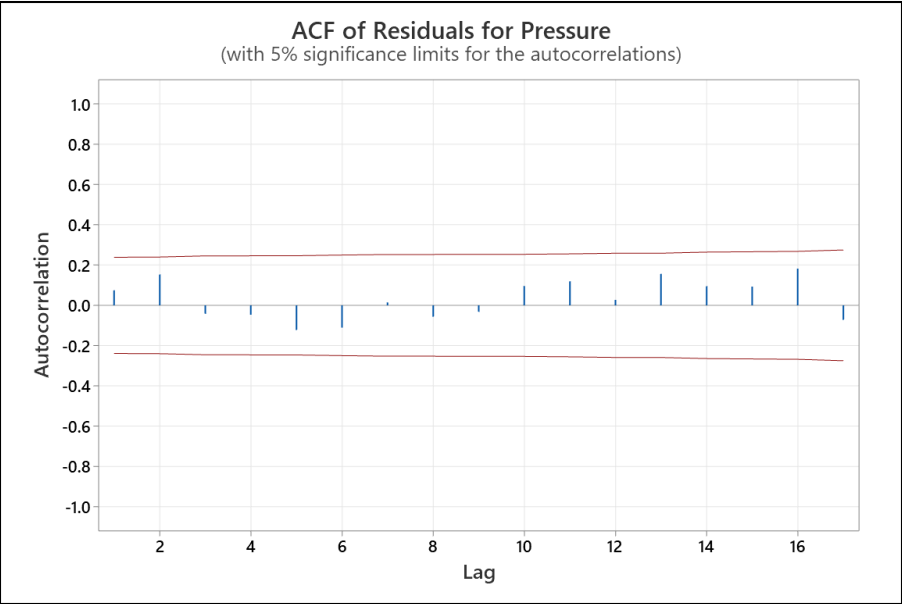
Final Estimates of Parameters

Type	Coef	SE Coef	T-Value	P-Value
AR 1	-0.426	0.116	-3.68	0.000
Constant	73.09	1.31	55.81	0.000
Mean	51.267	0.919		

The estimated model is: $X_t = 73.091 - 0.4257 X_{t-1} + e_t$

Exercise 1.1 Estimate the most suitable ARIMA model.

5) Check assumptions on residuals:



Normality test

Mean	-0.01488
StDev	10.87
N	70
RJ	0.988
P-Value	>0.100

Runs test

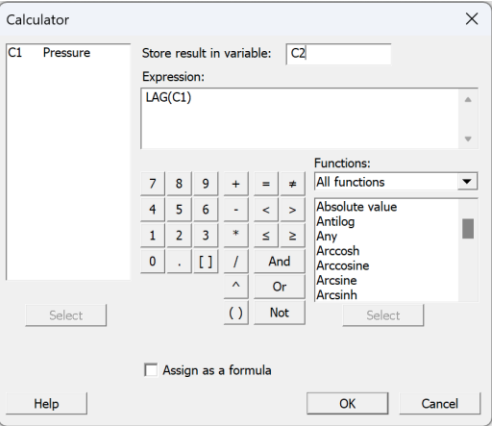
Number of Runs		
Observed	Expected	P-Value
37	35.97	0.804

Exercise 1.1 Estimate the most suitable ARIMA model.

6) Let's recompute the AR(1) model through linear regression, using as regressor the LAG1 of the series:

1 - First, store the LAG1 of the time series into a new column.

Data → Calculator



2 - Compute the regression model.

Stat → Regression → Regression → Fit regression model

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	73.09	6.14	11.90	0.000	
LAG1	-0.425	0.116	-3.65	0.001	1.00

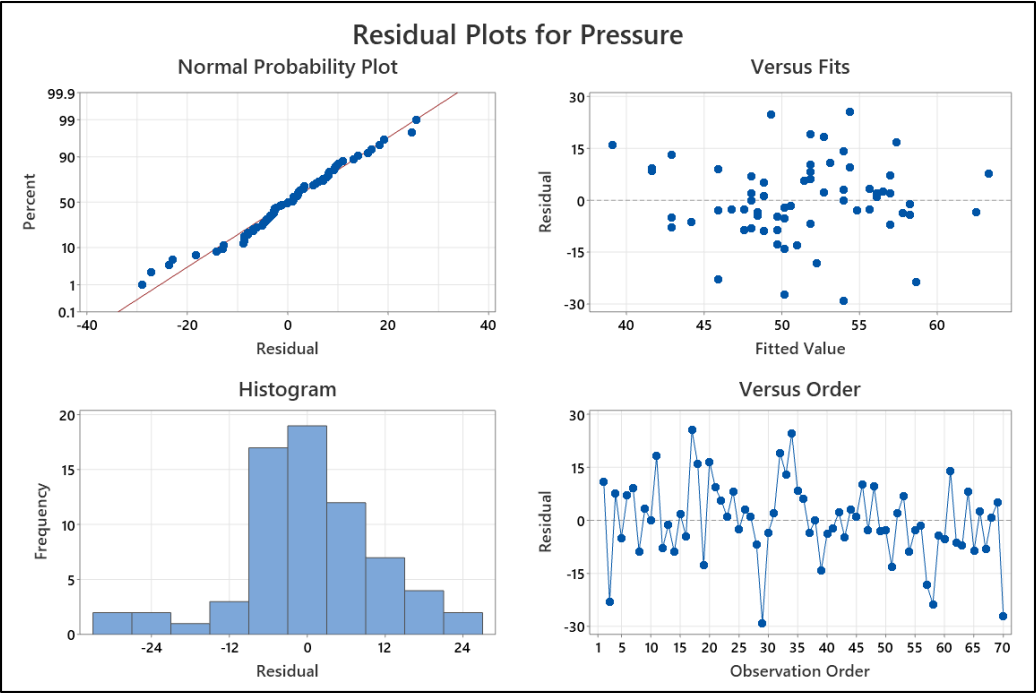
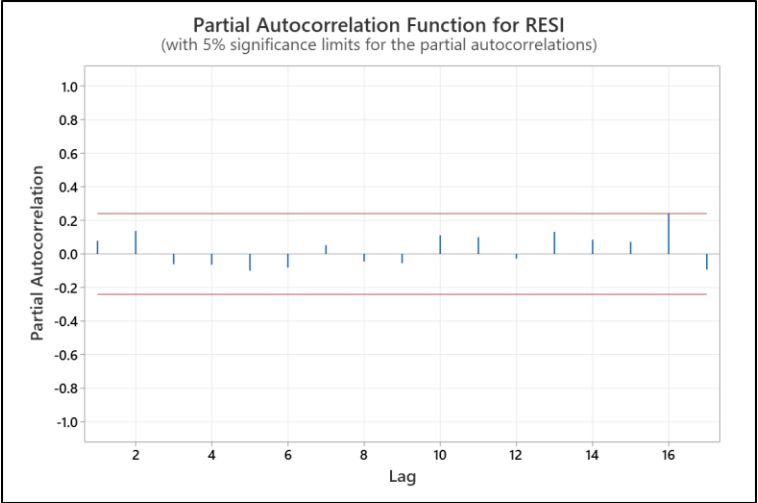
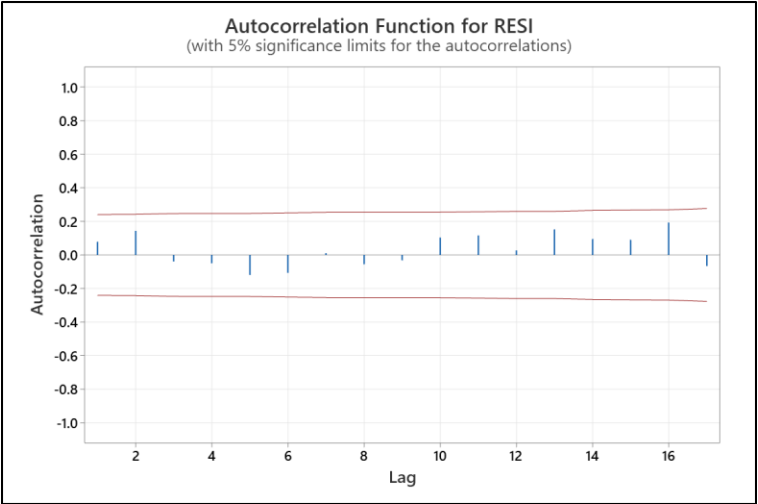
The estimated model is: $X_t = 73.0863 - 0.4249 X_{t-1} + e_t$

The coefficients of the AR(1) model fitted with ARIMA command are slightly different from the ones of the same model fitted with ordinary least squares because two different solvers are used (ARIMA uses log-likelihood).

Exercise 1.1 Estimate the most suitable ARIMA model.

6) Let's recompute the AR(1) model through linear regression, using as regressor the LAG1 of the series:

3 - Check assumptions on residuals.



Runs test

Test		
Null hypothesis H_0 : The order of the data is random		
Alternative hypothesis H_1 : The order of the data is not random		
Number of Runs		
<u>Observed</u>	<u>Expected</u>	<u>P-Value</u>
38	35.49	0.543

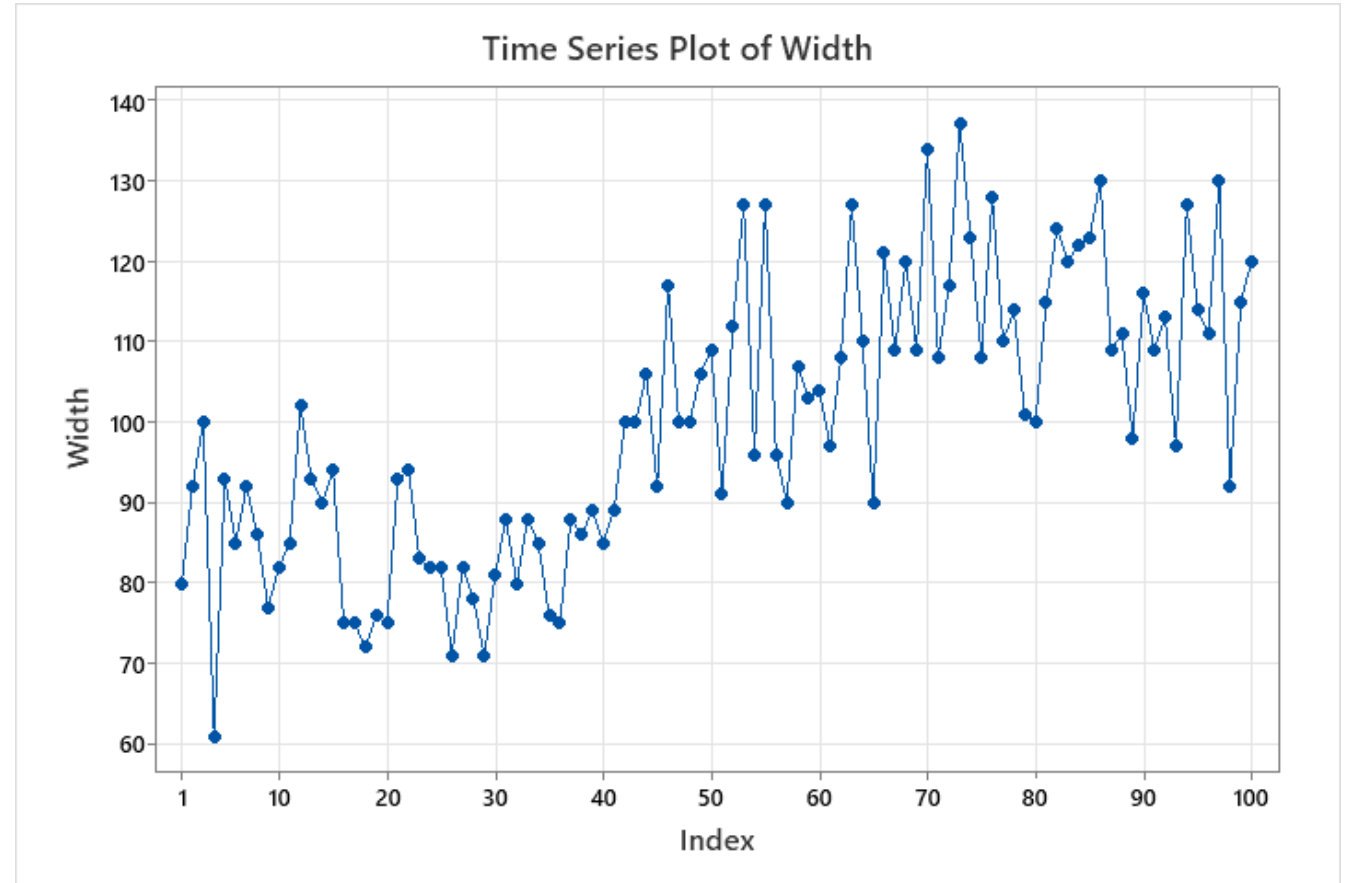
Normality test

Mean	-3.60420E-15
StDev	10.95
N	69
RJ	0.989
P-Value	>0.100

Exercise 2

In a process for the production of metal laminates we collected 100 sequential measurements of laminate width (*time series 'A'* “*Statistical Control by monitoring and feedback adjustment*” Box Luceño – J. Wiley)

1. Identify and fit a model for the data.



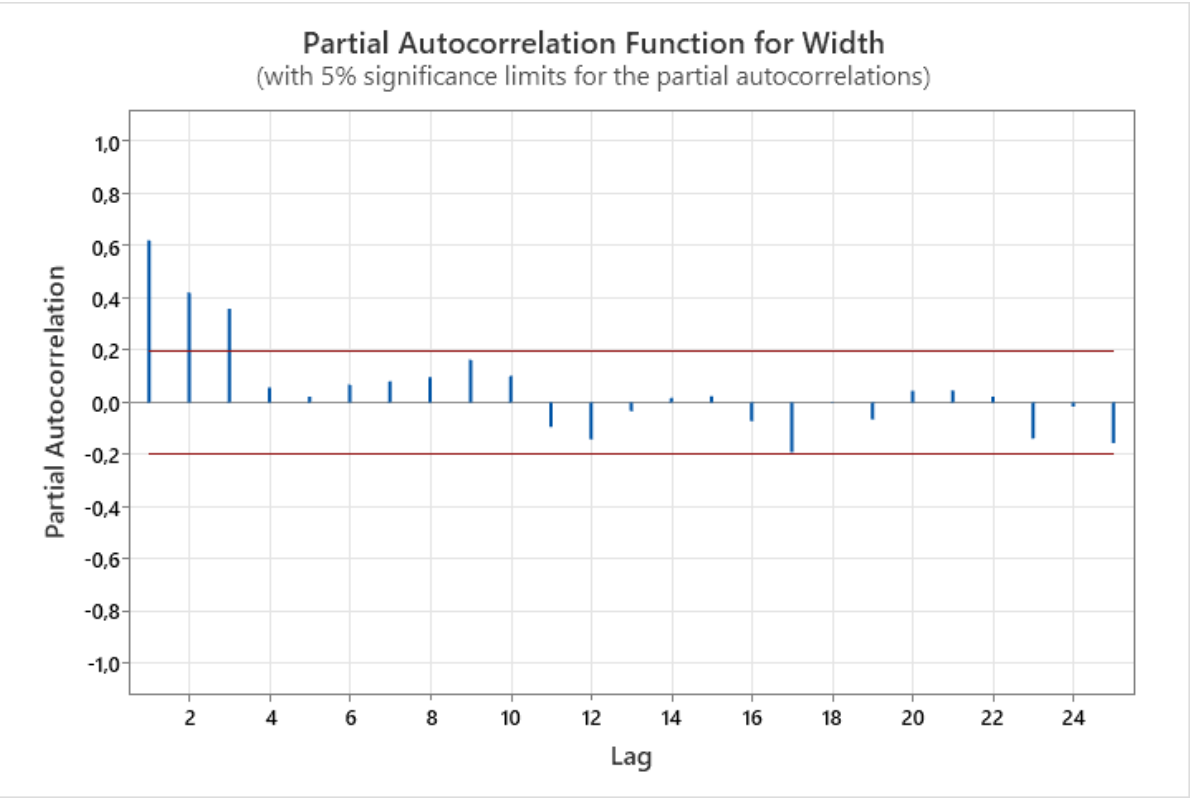
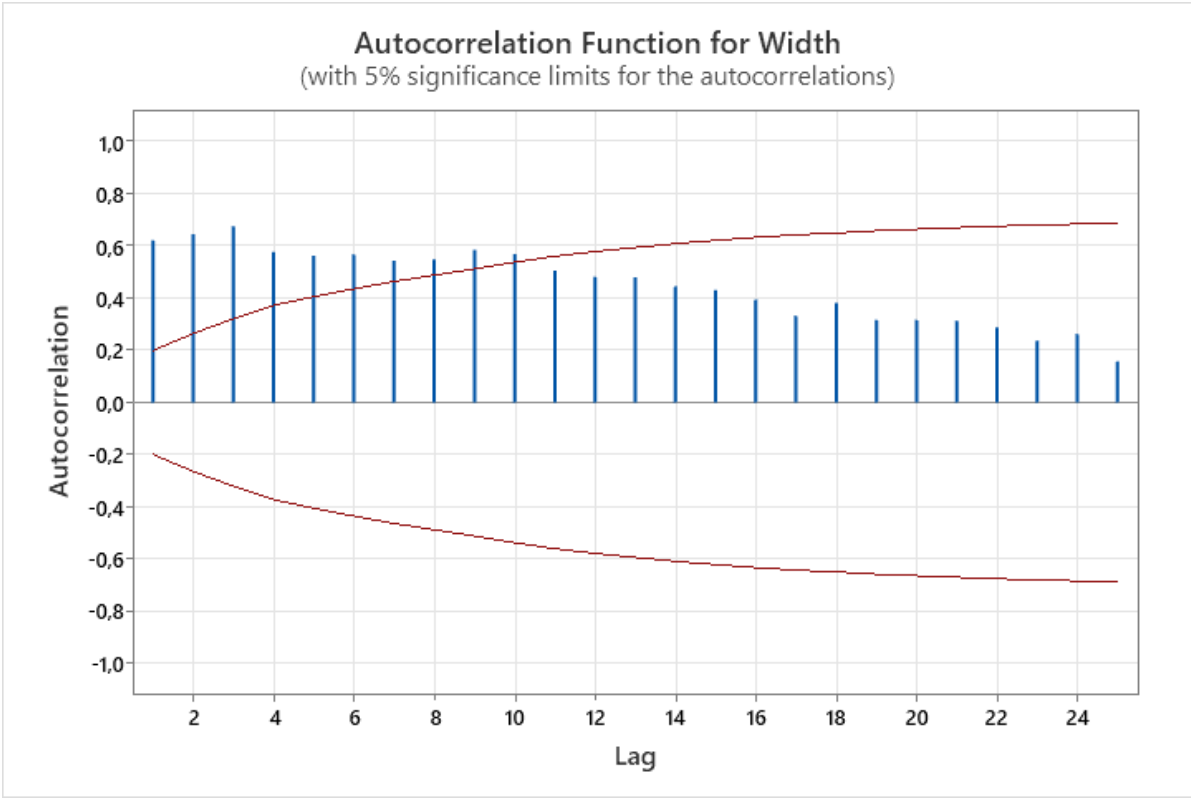
Exercise 2.1 Estimate the most suitable ARIMA model.

Let's verify the independence assumption with runs test and ACF/PACF:

Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
24	50,98	0,000

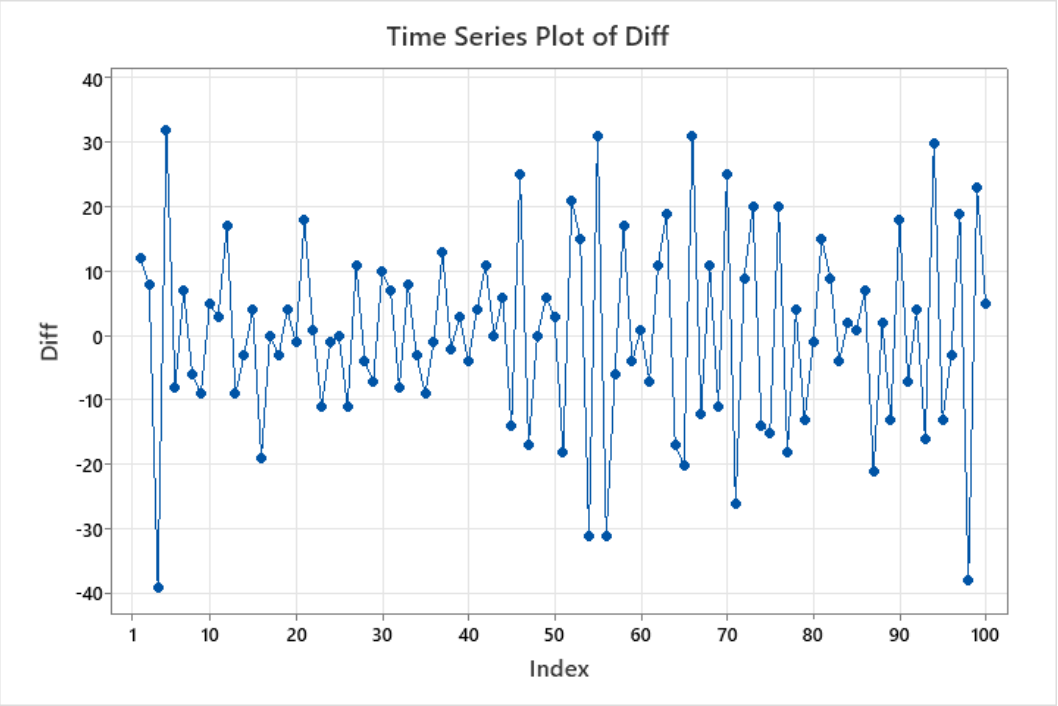


The process is **NON-STATIONARY**. Let's try to apply the difference operator.

Exercise 2.1 Estimate the most suitable ARIMA model.

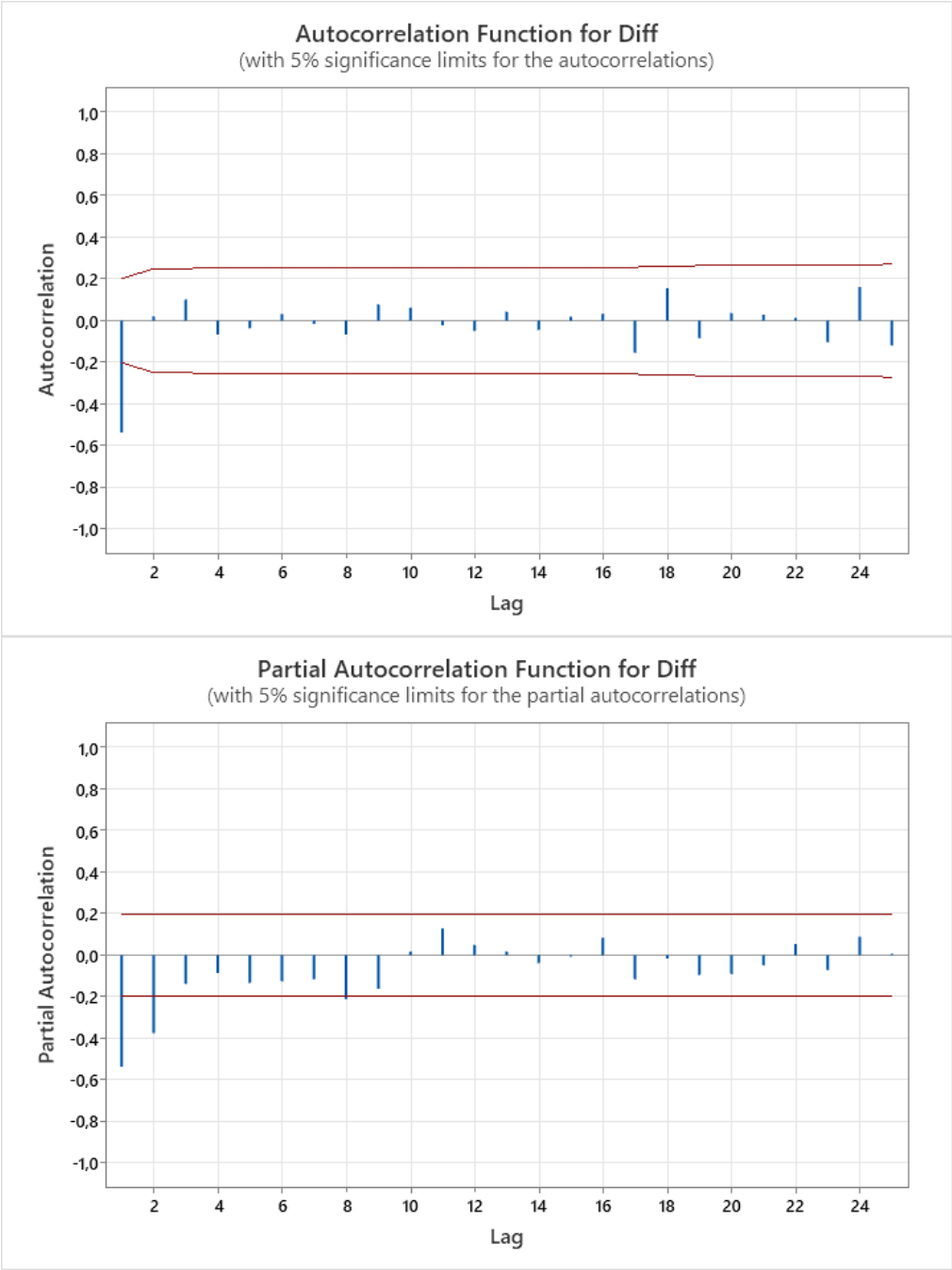
Let's try to apply the difference operator.

Stat → time series → differences



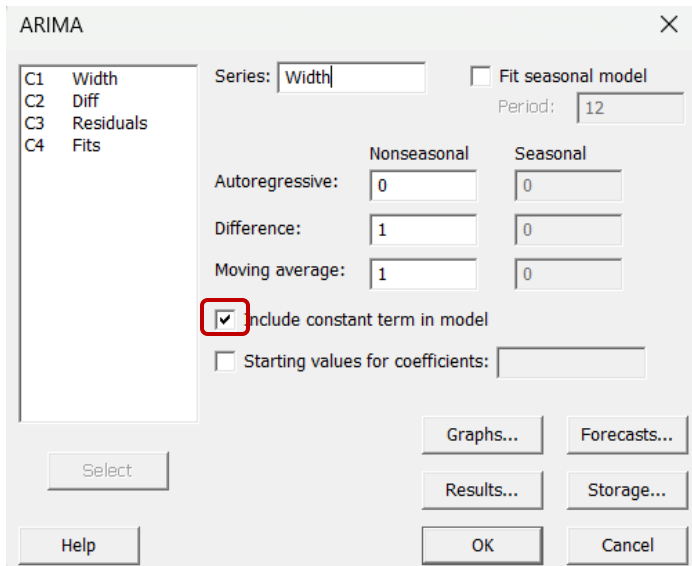
Runs test p-value on Diff = 0,00

After the differencing operation, the most suitable model seems to be an MA(1). Thus, the investigated model is **ARIMA(0,1,1)**



Exercise 2.1 Estimate the most suitable ARIMA model.

Stat → time series → ARIMA



The ARIMA dialog box shows the following settings:

- Series: Width
- Fit seasonal model: ☐
- Period: 12
- Autoregressive: Nonseasonal 0, Seasonal 0
- Difference: Nonseasonal 1, Seasonal 0
- Moving average: Nonseasonal 1, Seasonal 0
- ☒ Include constant term in model
- ☐ Starting values for coefficients:

Buttons at the bottom: Select, Help, Graphs..., Forecasts..., Results..., Storage..., OK, Cancel.

Final Estimates of Parameters

Type	Coef	SE	Coef	T-Value	P-Value
MA 1	0,8225	0,0573	14,35	0,000	
Constant	0,313	0,205	1,53	0,130	

The computed ARIMA model is in the form:

$$Y_t - Y_{t-1} = \nabla Y_t = \mu - \theta_1 \epsilon_{t-1} + \epsilon_t$$

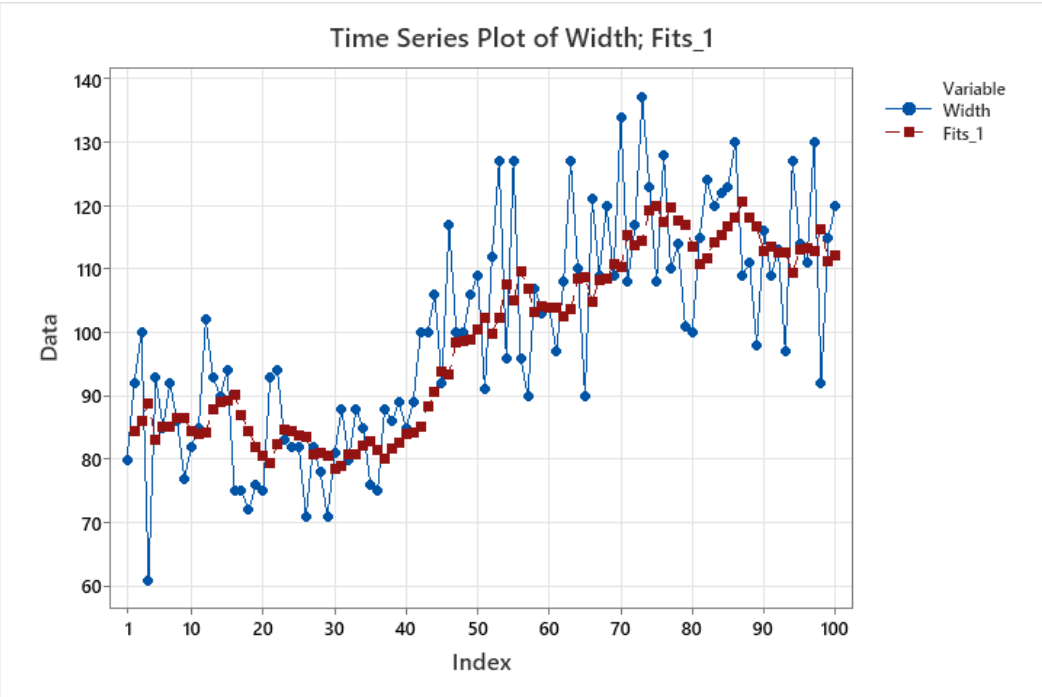
The constant term has a p-value of 0.13.
Let's remove the constant value by omitting the constant parameter.

Final Estimates of Parameters

Type	Coef	SE	Coef	T-Value	P-Value
MA 1	0,7916	0,0610	12,98	0,000	

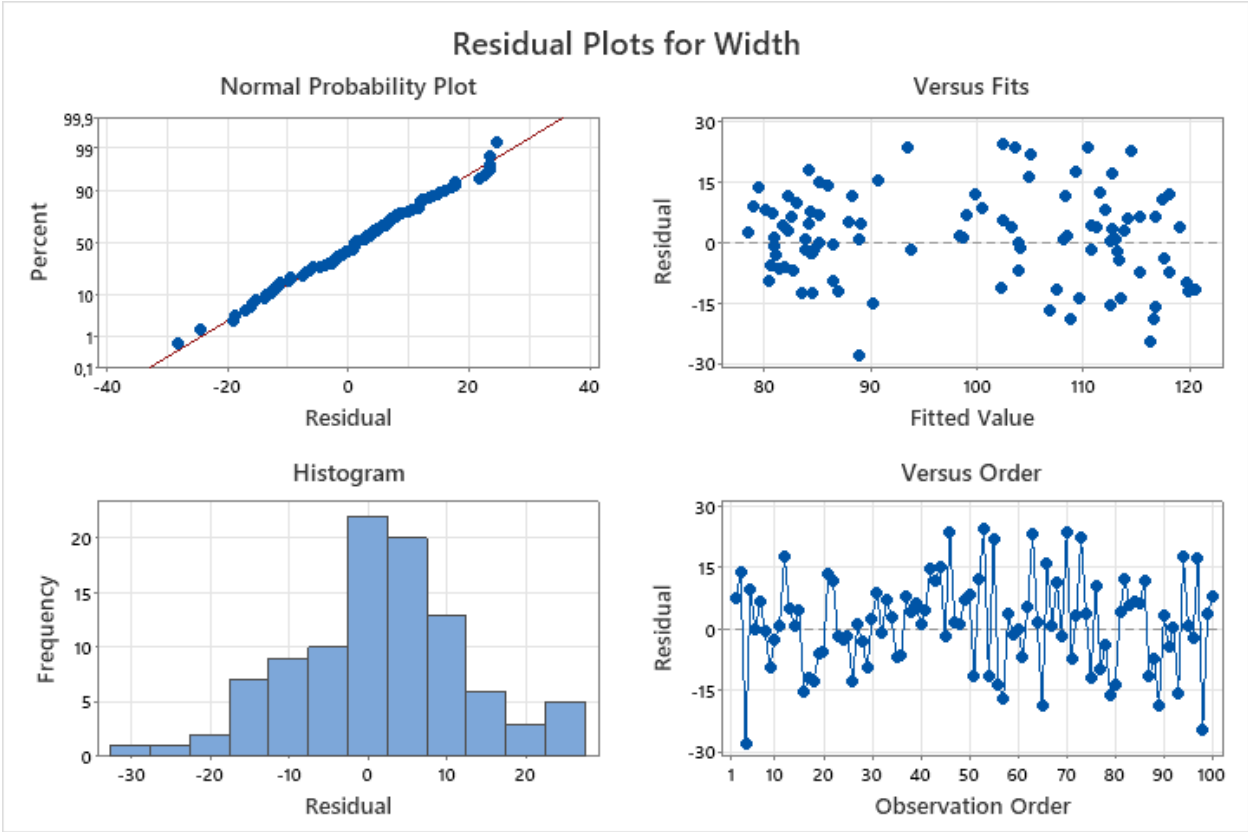
The computed model is in now the form:

$$Y_t - Y_{t-1} = \nabla Y_t = \theta_1 \epsilon_{t-1} + \epsilon_t$$



Exercise 2.1 Estimate the most suitable ARIMA model.

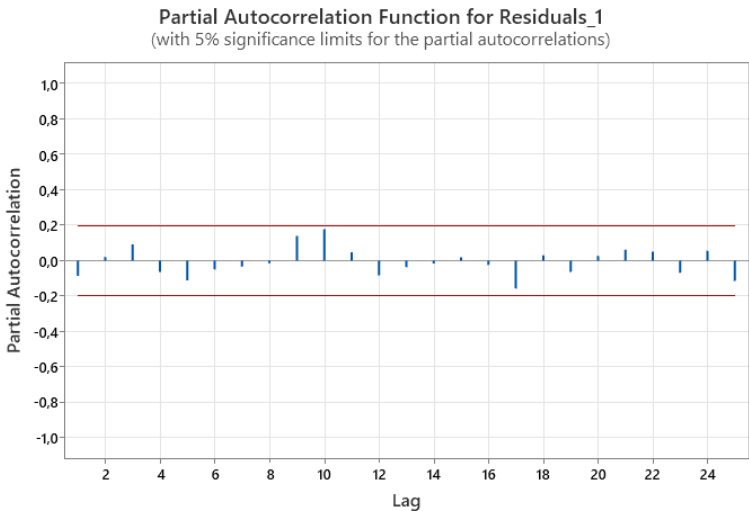
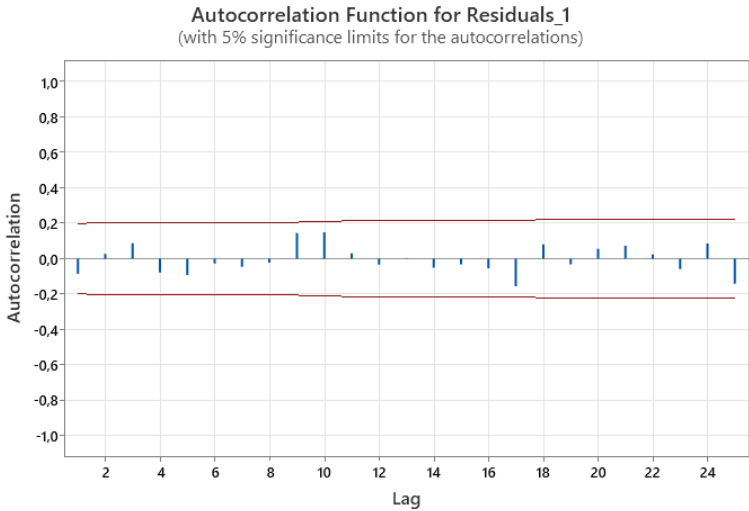
Let's check the assumptions on the residuals:



Runs test:

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
51	50,49	0,919



Shapiro-Wilk p-value on Residuals > 0,1

The model is adequate.